

discount1, discount2, parking charge, price $\leftarrow 0$

OUTPUT "Enter Vehicle type - (E)lectric or (H)ybrid?"

INPUT vtype

OUTPUT "Current battery percentage?"

INPUT cur-battery

OUTPUT "Required charging level?"

INPUT req-battery

OUTPUT "Enter expected parking duration"

INPUT park-duration

OUTPUT "Enter current time in 24-hour format"

INPUT current-time

OUTPUT "Have parking membership? (Y or N)"

INPUT membership

OUTPUT "Do you have disabled-person priority status?"

INPUT disabled-status

OUTPUT "Is charging station available?"

INPUT station-available

charging ← FALSE

Charging ← FALSE

IF station-available = 'Y':

~~charging~~

IF v-type = 'H' AND cur-battery ≥ 40 :

OUTPUT "Vehicle doesnot qualify for EV charging"

ELSE: ~~valid ← TRUE~~ ~~Pending ← TRUE~~

~~charging ← TRUE~~

req-charging ← req-battery - cur-battery

IF req-charging $\leq$ cur-battery:

OUTPUT "No charging required"

ELSE: ~~charging ← TRUE~~

IF cur-battery ≤ 15 AND req-battery ≥ 80 :

priority ← "Emergency Charging priority"

ELSE IF (cur-battery ≤ 30) AND (disabled-status = 'Y')
OR membership = 'Y':

priority ← "Priority charging"

ELSE:

priority ← "Normal Charging"

ENDIF

ENDIF

ENDIF

ELSE:

IF v-type = 'H':

OUTPUT "Charging unavailable - Parking only"

ELSE:

OUTPUT "No charging slot available"

ENDIF

ENDIF

IF charging = TRUE:

IF current_time < 17:00 OR current_time > 22:00:

IF membership = 'Y'

discount1 ← 20

price ← 35

ELSE IF current_time > 17:00 AND current_time < 22:00:

IF membership = 'Y'

discount1 ← 10

price ← 50

ENDIF

IF priority = 'Emergency Charging priority'

discount1 ← 0

ENDIF

ENDIF

IF park_duration ≤ 2:

park_charge ← 200

ELSE IF park_duration ≤ 5:

park_charge ← 400

ELSE: ~~IF park_duration~~

park_charge ← 700

ENDIF

IF membership = 'Y':

discount2 ← 20

ELSE IF

disabled_status = 'Y':

discount2 ← 100

ENDIF

Task 6

```
IF park_duration > 8:  
    OUTPUT "Long-stay warning. Please relocate your vehicle  
        after charging"
```

```
ELSE:
```

```
    OUTPUT "Standard parking duration"
```

```
OUTPUT "Vehicle type:", v_type
```

```
OUTPUT "Current battery percentage:", cur_battery
```

```
OUTPUT "Required battery percentage:", req_battery
```

```
OUTPUT "Charging priority", priority
```

```
IF charging < TRUE:
```

```
    OUTPUT "Charging priority:", priority
```

```
    OUTPUT "Charging cost:", price
```

```
ENDIF
```

```
    OUTPUT "Charging discount:", discount1
```

```
ENDIF
```

```
IF
```

```
OUTPUT "Parking cost", park_charge
```

```
OUTPUT "Parking discount", discount2
```

```
total_cost <- (price - (price * (discount1 / 100))) + (park_charge -  
                                                    (park_charge * (discount2 /  
                                                    100)))
```

```
OUTPUT "Total cost:", total_cost
```


IPAC

INPUT	Required results
v_type cur_battery req_battery park_duration current_time membership disabled_status station_available	Vehicle type Current battery percentage required battery % Charging priority charging cost parking cost discount 1 discount 2 total_cost

Processing required	Alternative solution
discount 1 = 0 discount 2 = 0 park_charge = 0 price = 0 charging_cost = price - (price * discount 1 / 100) parking_cost = park_charge - (park_charge * discount 2 / 100) total_cost = charging_cost + parking_cost	set outputs as constants

IPO

INPUT	Processing	OUTPUT
v_type cur_battery req_battery park_duration current_time membership disabled_status station_available	1. Validate v_type 2. Calculate energy $req_charging = req_battery - cur_battery$ 3. Priority ^{calculate} 4. price discount 1 discount 2 5. $charging_cost = price - (price * discount 1 / 100)$ 6. parking cost = $park_cost - (park_cost * discount 2 / 100)$	v_type cur_battery req_battery priority peak/off-peak disc. total_cost

Start

Date: _____

discount 1 $\leftarrow$ 0
discount 2 $\leftarrow$ 0
park_charge $\leftarrow$ 0
price $\leftarrow$ 0

~~Parking~~Charging $\leftarrow$ FALSE

/ OUTPUT "Enter Vehicle type: (E)lectric or (H)ybrid" /

/ INPUT v_type /

/ OUTPUT "Current battery percentage:" /

/ INPUT cur_battery /

/ OUTPUT "Required charging level:" /

/ INPUT ~~cur~~ req_battery /

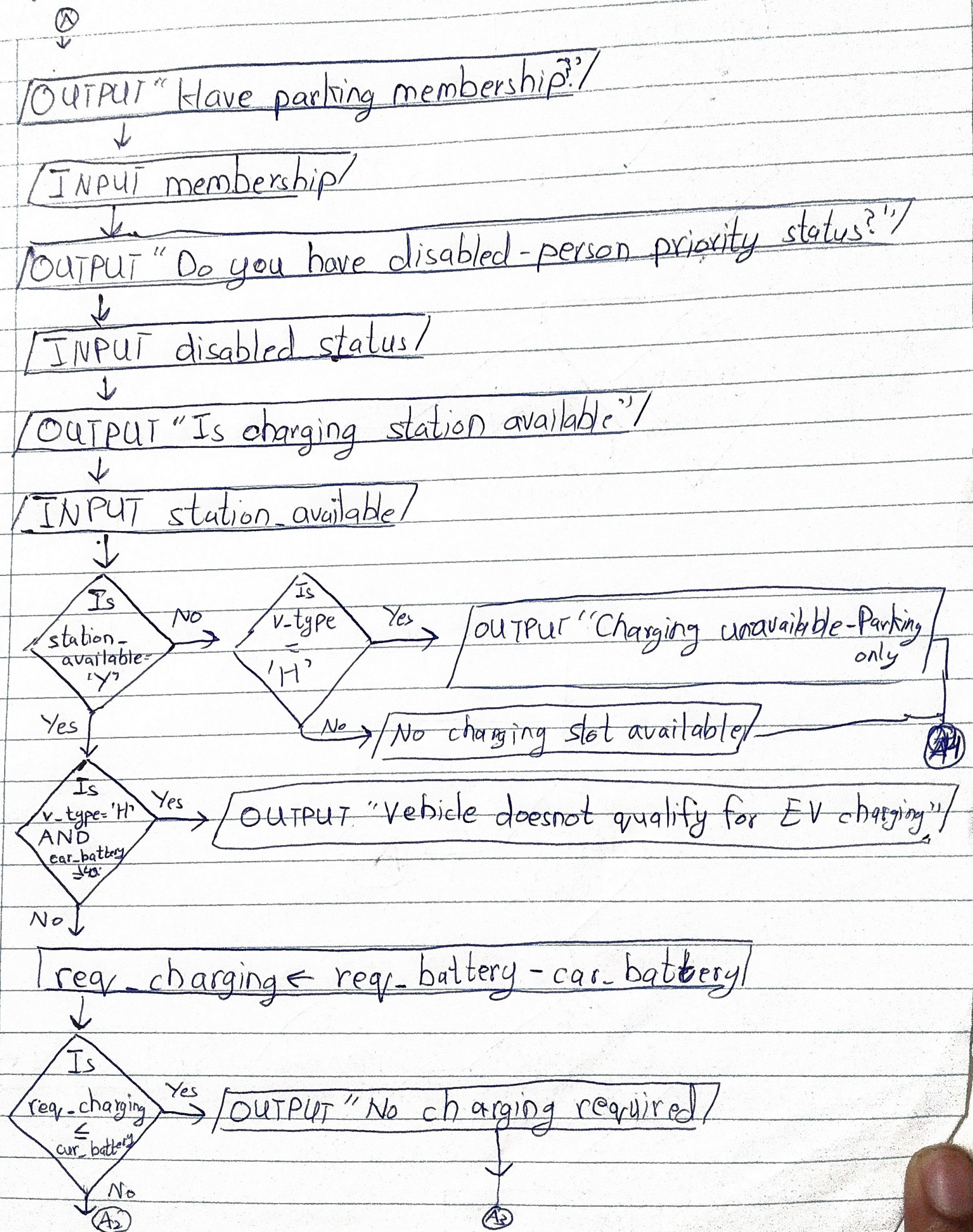
/ OUTPUT "Enter expected parking duration" /

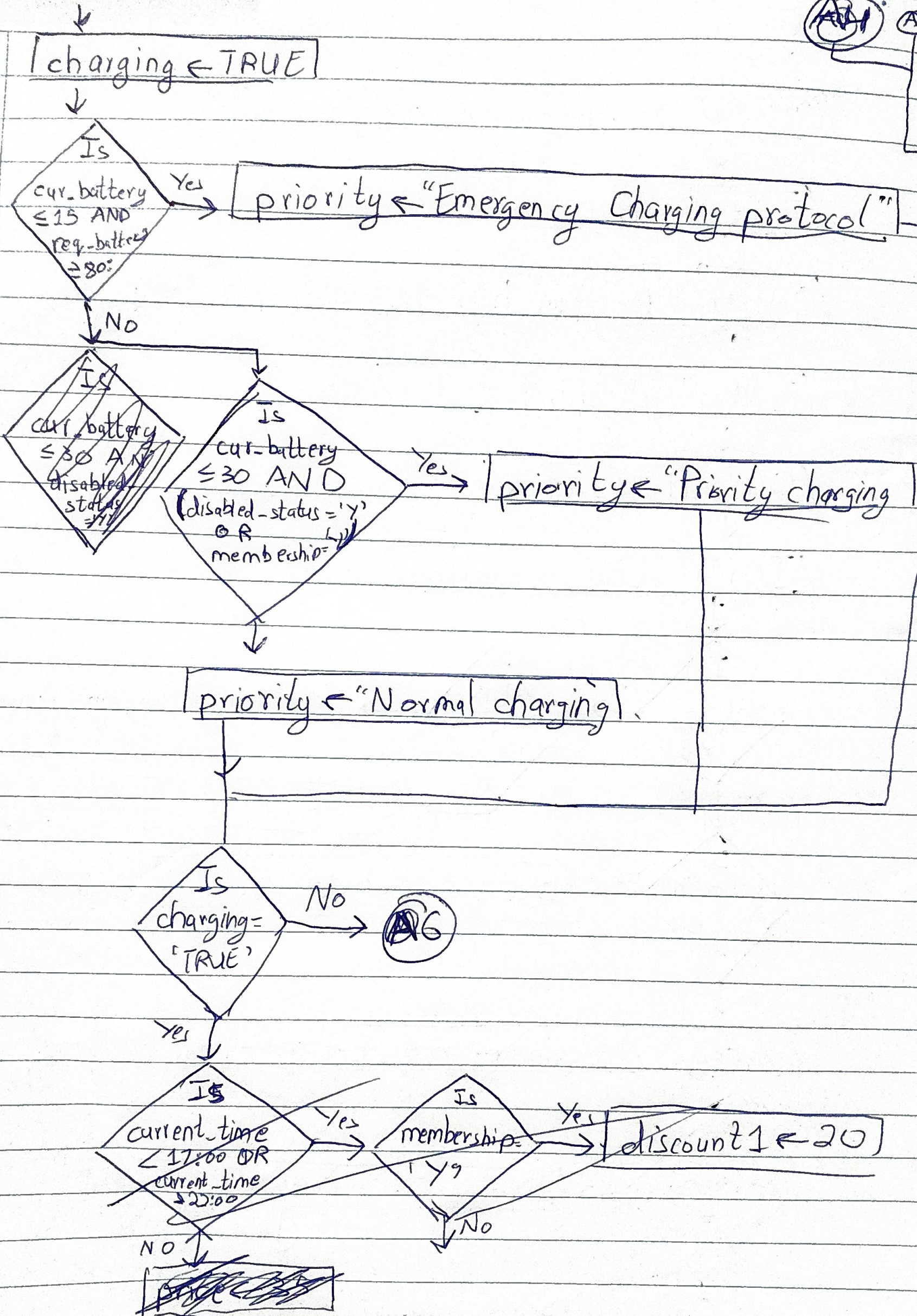
/ INPUT park_duration /

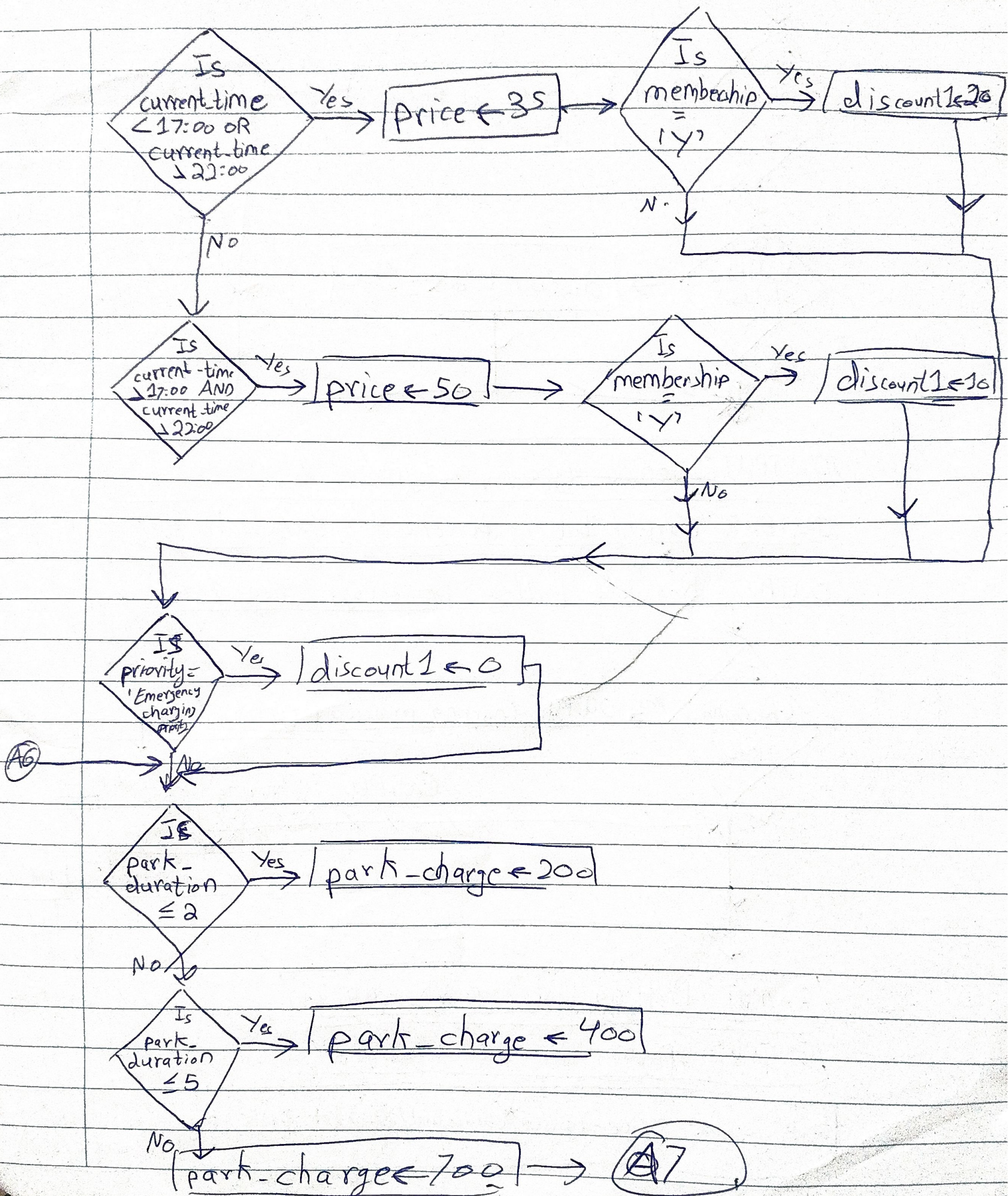
/ OUTPUT "Enter current time (24 hour format)" /

/ INPUT current_time /

(A)

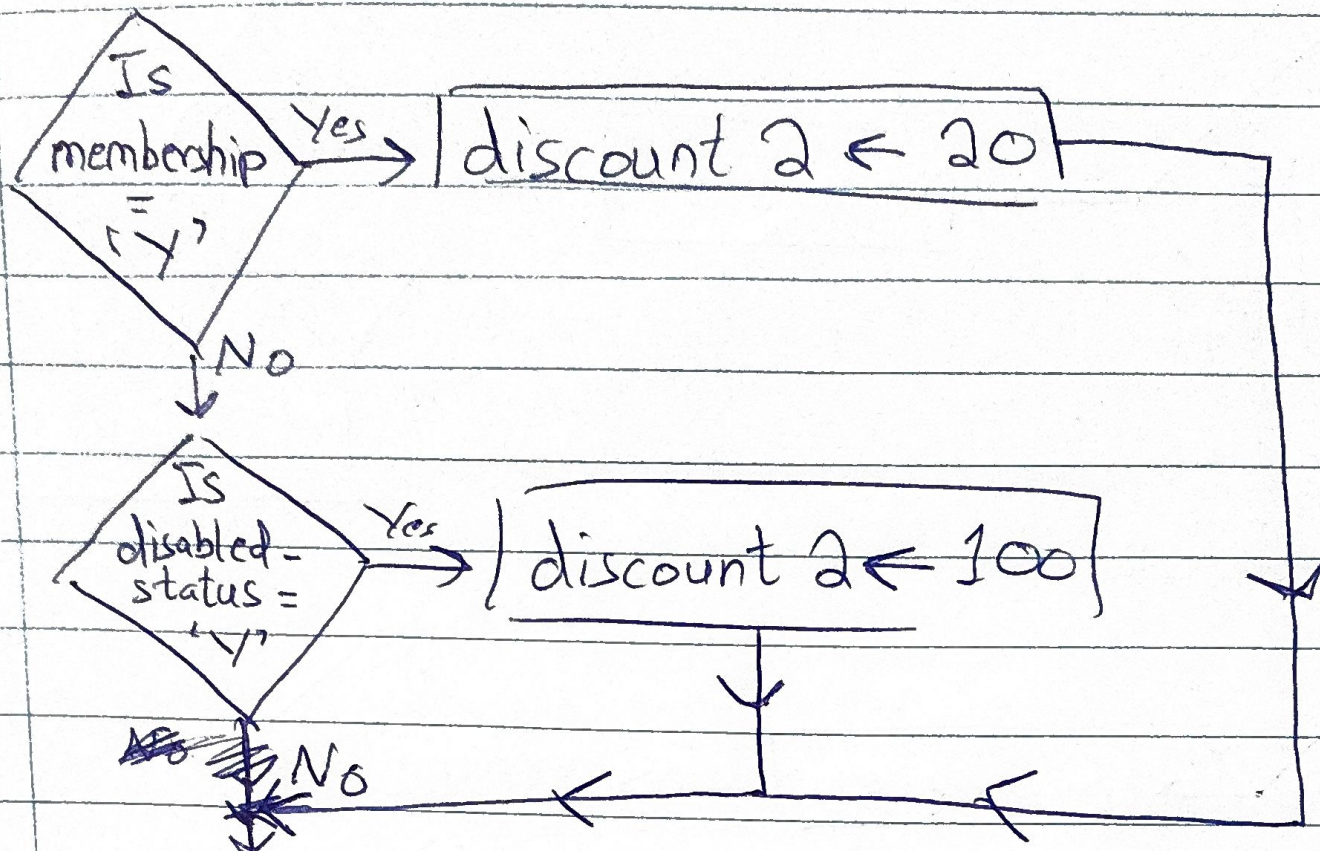






A7

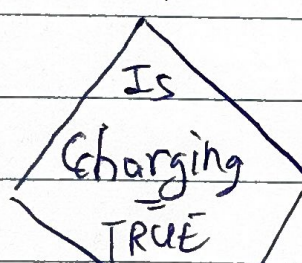
Date: _____



OUTPUT "Vehicle type", v_type

OUTPUT "Current battery percentage", cur_battery

OUTPUT "Required battery percentage", req_battery



Yes → OUTPUT "Charging priority", priority

OUTPUT "Charging cost", ^{price}cost

OUTPUT "Charging discount", discount1

OUTPUT "Parking cost", park_charge

OUTPUT "Parking discount", discount2

total_cost ← price - (price * (discount1 / 100)) + park_charge - (park_charge * (discount2 / 100))

OUTPUT "Total cost", total_cost